

pler itself has been known and used for a very long time but it has never been used and applied at this high a frequency. Since we are sampling the signal right after the LNA, the sampler needs to operate at very, very high frequencies. In one of our products, we sometimes sample the signal at more than 3 GHz. So a lot of funny signal processing happens at 3 GHz. That's not a joke; the DSP-RF has quite a bit of complicated circuitry to deal with.

The ideal processing procedures regarding the circuit topologies took us a couple of generations to get right. In essence, the design of a sampler might seem easy but to get it working right is a different ball-game altogether. While trying to better the performance of the sampler, there are some critical characteristics that you have to maintain. Also, the sampling should not distort the signal. Then you have specific measures in RF that describe what that means, in numbers.

Could you please elaborate on these measures?

Those measures are noise figures, which describe how much noise you are adding in the circuit. The second is linearity, which is measured by many different parameters, IN2, IN3, second-order and third-order distortion. Getting all these parameters to specifications simultaneously is an extremely difficult challenge. Our first circuit design was good for noise reduction but not so much for maintaining linearity. The second one, on the other hand, was good for inter-modulation but not so much for noise reduction. So getting the ideal circuit topology and a compromise in performance between the two designs is what took the most work and trips to the drawing board.

What technologies are being adopted as low-power wireless gains prominence?

When you say low-power wireless, we are including transmitters (unlike the technologies which we have discussed till now) and transceiver technologies. Again, this is another area where DSP-RF shines because by using DSP-RF we are able to do what is called 'polar modulation.' Polar modulation is a way to do very efficient transmitter design.

In most transceivers, the transmitter ends up being the most power-hungry device. Polar modulation allows us to optimise the power consumption of the transmitter. The problem, of course, is in implementing polar modulation: You get into a lot of technological roadblocks, which other people haven't been able to overcome yet—especially as you get to the wide-band polar modulation. When you do that, DSP-RF shines and compares very well with ideal polar modulation. Actually, it is part of our roadmap to add wide-band polar modulation to our product mix. In the coming IoT era, polar modulation is very important.

How is implementing IPs in SoCs made easier with DSP-RF?

Since DSP-RF is inherently digital in nature, getting

that to work with a large digital baseband on the same die, which is what we call an SoC, by definition, is much easier. Let us say you want to make a WLAN SoC. There is quite a bit of pain that has to be overcome in a standard design, integrating the analogue section, RF section and the big digital baseband processing section. Since the RF section is itself so digital-centric, it is much easier to integrate the technology using a digital setup, which is what the DSP-RF is.

One of the big requirements in TV tuners is that the spurious input be very low. By spurious input, we are referring to the amount of unwanted signals that end up at the antenna port. When you design with standard analogue technology, it is very difficult to get low spurious input because mixing and the subsequent signal processing have quite a few non-linearities in them. But when you use a sampling approach, if you are clever about picking the right frequencies, the spurious input can be avoided completely and you actually get a very high-performance tuner with respect to the spurious input.

Speaking of tuner design, what are the challenges faced while designing a universal tuner? Could you give an example too?

Every region has its own standards. Analogue standards are PAL, NTSC and SECAM. China has a new digital standard coming up, Japan has ISDB-T, Europe has DVB-T, the US has ATSC and so on. It almost feels like an alphabet soup of different standards, with each standard having its own requirements.

Digital standards have their own requirements and the analogue ones are also extremely tricky because in analogue the human eye will catch even the smallest impairment in the video. For example, even if a tiny amount of spurious input, say, a 1 μ V signal, is applied at the LNA input port, you can see the distortion in an analogue TV. So each standard has its own set of challenges that you have to solve. Getting all that to work simultaneously, on a single die, is extremely tricky and the most challenging in this line of work.

Finally, could you detail the challenges faced with DSP-RF in this domain?

Every aspect has had its challenges. Both architecture design, which we talked about earlier, as well as implementing the architecture in ICs posed challenges in the sense that since this hasn't been done before, getting the right circuit topologies to work was not a first-time success. It required multiple iterations.

By now, we understand DSP-RF a lot more than we did when we started the company. We are now able to appreciate what is possible and what isn't in IC design, and the only way to gain that appreciation is by actually going through the process of implementing it. ●

(The complete interview can be read at electronicsforu.com)